

# Kinematic Analysis and Trajectory Execution of the KUKA LBR MED Surgical Manipulator

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# 1 Denavit–Hartenberg Convention

## 1.1 Robot Description

The KUKA LBR Med 7 R800 is a 7-DOF redundant surgical manipulator with all revolute joints. Its kinematic structure consists of alternating perpendicular joint axes, giving one degree of intrinsic redundancy over the 6-DOF task space.

Table 1: Link dimensions.

| Parameter | Value   | Description                              |
|-----------|---------|------------------------------------------|
| $d_1$     | 0.340 m | Base platform to shoulder (joint 2 axis) |
| $d_3$     | 0.400 m | Shoulder to elbow (joint 4 axis)         |
| $d_5$     | 0.400 m | Elbow to wrist centre (joint 6 axis)     |
| $d_7$     | 0.126 m | Wrist centre to flange                   |

## 1.2 Frame Assignment (Standard DH Convention)

Following the standard DH procedure:

1. Number joint axes  $z_0, \dots, z_6$  where  $z_i$  is the axis of joint  $i+1$ .
2. Place Frame 0 at the base with  $z_0$  pointing vertically upward.
3. For  $i = 1, \dots, 6$ , place  $O_i$  at the intersection of  $z_{i-1}$  and  $z_i$ .
4. Define  $x_i$  along the common normal from  $z_{i-1}$  to  $z_i$ .
5. Define  $y_i$  to complete the right-handed frame.
6. Place Frame 7 at the end-effector with  $z_7 \parallel z_6$  and  $O_7$  at the flange centre.

**Key structural observation:** all consecutive joint axes intersect, so all link offsets satisfy  $a_i = 0$ .

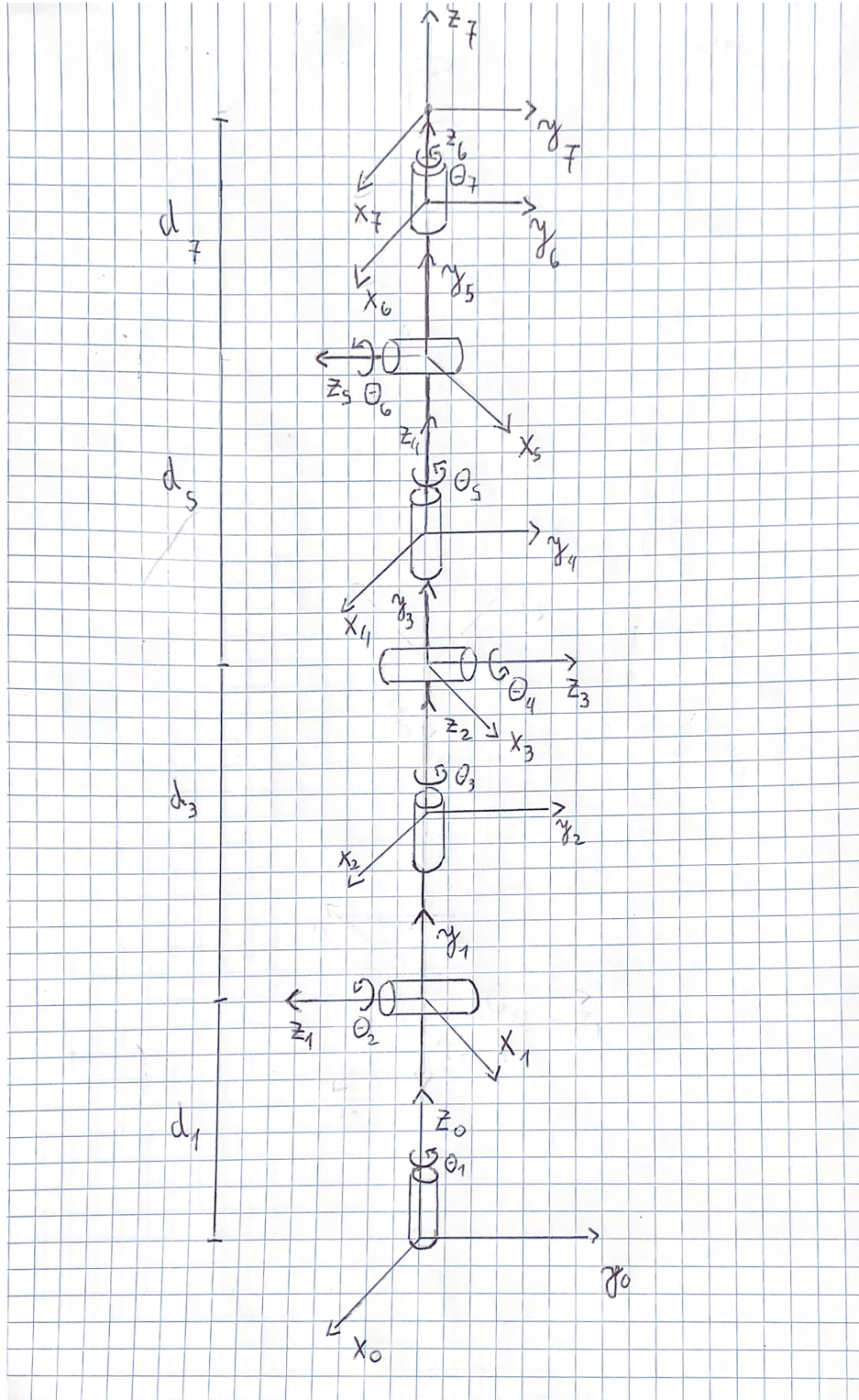


Figure 1: Hand-drawn DH frame assignment of the KUKA LBR MED.

### 1.3 DH Parameter Table

The homogeneous transformation per link is:

$$A_i^{i-1}(q_i) = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i) \quad (1)$$

Table 2: Standard DH parameters for the KUKA LBR MED.

| Link $i$ | $a_i$ (m) | $\alpha_i$ (rad) | $d_i$ (m) | $\theta_i$ | Joint                 |
|----------|-----------|------------------|-----------|------------|-----------------------|
| 1        | 0         | $+\pi/2$         | 0.340     | $q_1$      | Base rotation         |
| 2        | 0         | $-\pi/2$         | 0         | $q_2$      | Shoulder tilt         |
| 3        | 0         | $-\pi/2$         | 0.400     | $q_3$      | Upper-arm rotation    |
| 4        | 0         | $+\pi/2$         | 0         | $q_4$      | Elbow tilt            |
| 5        | 0         | $+\pi/2$         | 0.400     | $q_5$      | Forearm rotation      |
| 6        | 0         | $-\pi/2$         | 0         | $q_6$      | Wrist tilt            |
| 7        | 0         | 0                | 0.126     | $q_7$      | End-effector rotation |

## 2 Direct Kinematics

### 2.1 Method

Using the Symbolic Robotics MATLAB Toolbox, the direct kinematics transformation is the ordered product of link matrices:

$$T_7^0(q) = A_1^0(q_1) A_2^1(q_2) \cdots A_7^6(q_7) \quad (2)$$

The position  $p$  and rotation matrix  $R$  of the end-effector in the base frame are extracted from  $T_7^0$  as its upper-left  $3 \times 3$  block and upper-right  $3 \times 1$  column, respectively.

### 2.2 Simulink Model

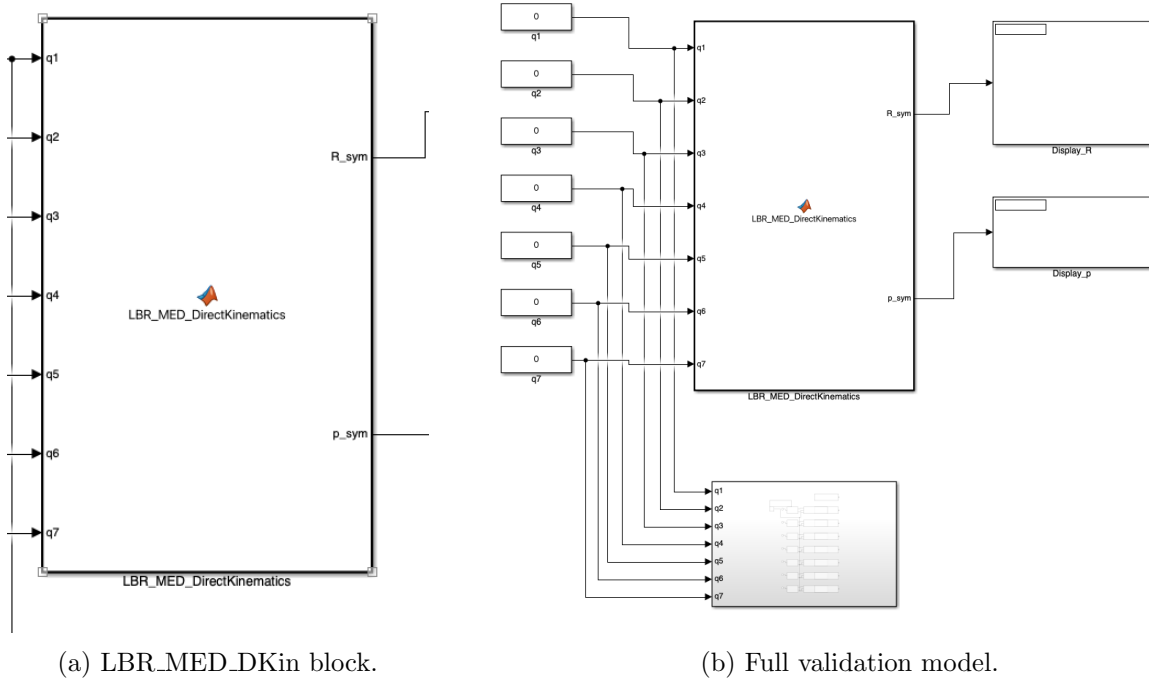


Figure 2: Direct kinematics Simulink model (LBR\_MED\_Simul.slx).

### 2.3 Validation

Four configurations are chosen where the end-effector pose can be verified by geometric reasoning alone.

Table 3: Direct kinematics validation results.

| Config           | $q$ (rad)                      | Expected $p$ (m)          | Expected $R$  | $\ p_{\text{err}}\ $ |
|------------------|--------------------------------|---------------------------|---------------|----------------------|
| 1: Home          | all = 0                        | $[0, 0, 1.266]^\top$      | $I_3$         | $< 10^{-12}$         |
| 2: Shoulder bend | $q_2 = \pi/2$ , rest = 0       | $[-0.926, 0, 0.340]^\top$ | $R_y(-\pi/2)$ | $< 10^{-12}$         |
| 3: Base spin     | $q_1 = \pi/2$ , rest = 0       | $[0, 0, 1.266]^\top$      | $R_z(\pi/2)$  | $< 10^{-12}$         |
| 4: Elbow fold    | $q_2 = q_4 = \pi/2$ , rest = 0 | $[-0.400, 0, 0.866]^\top$ | $I_3$         | $< 10^{-12}$         |

All four configurations pass with machine-precision errors, confirming the symbolic DH product and the MATLAB function handles are correct.

### 3 Inverse Kinematics (Closed-Form)

#### 3.1 Kinematic Decoupling

The robot has a spherical wrist (joints 5–7, axes  $z_4, z_5, z_6$  intersecting at  $O_6$ ), enabling the classical decoupling:

- **Arm problem** (joints 1–4): position the wrist centre  $O_6$ .
- **Wrist problem** (joints 5–7): orient the end-effector.

**Redundancy resolution:** fixing  $q_3 = 0$  reduces the arm to three unknowns ( $q_1, q_2, q_4$ ) and yields a unique closed-form solution per branch.

#### 3.2 Closed-Form Solution

##### Step 1 — Wrist centre

$$p_W = p_e - d_7 a_e, \quad a_e = R_d[:, 3] \quad (3)$$

##### Step 2 — Base rotation $q_1$

$$q_1 = \text{atan2}(p_{Wy}, p_{Wx}) \quad \text{or} \quad q_1 + \pi \quad (\text{back-arm branch}) \quad (4)$$

##### Step 3 — Elbow angle $q_4$ (law of cosines)

With  $X = p_{Wx} \cos q_1 + p_{Wy} \sin q_1$  and  $Z = p_{Wz} - d_1$ :

$$\cos q_4 = \frac{X^2 + Z^2 - d_3^2 - d_5^2}{2d_3d_5}, \quad \sin q_4 = \pm \sqrt{1 - \cos^2 q_4} \quad (5)$$

##### Step 4 — Shoulder angle $q_2$

$$\begin{bmatrix} -(d_3 + d_5 c_4) & d_5 s_4 \\ d_5 s_4 & d_3 + d_5 c_4 \end{bmatrix} \begin{bmatrix} s_2 \\ c_2 \end{bmatrix} = \begin{bmatrix} X \\ Z \end{bmatrix} \Rightarrow q_2 = \text{atan2}(s_2, c_2) \quad (6)$$

##### Steps 5–7 — Wrist angles $q_5, q_6, q_7$

$$R_{47} = (R_0^4)^\top R_d \quad (7)$$

Reading off entries of the product  $A_5 A_6 A_7$ :

$$q_6 = \text{atan2}\left(\sqrt{1 - R_{47}(3, 3)^2}, R_{47}(3, 3)\right) \quad (8)$$

$$q_5 = \text{atan2}(-R_{47}(2, 3), -R_{47}(1, 3)) \quad (\sin q_6 \neq 0) \quad (9)$$

$$q_7 = \text{atan2}(-R_{47}(3, 2), R_{47}(3, 1)) \quad (10)$$

The solver generates up to **8 closed-form solutions** (2 base  $\times$  2 elbow  $\times$  2 wrist branches) and returns the one closest to a supplied reference configuration  $q_{\text{prev}}$ .

### 3.3 Singularity Conditions

Table 4: IK singularity conditions.

| Label | Condition             | Geometric meaning                                   |
|-------|-----------------------|-----------------------------------------------------|
| S1    | $p_{Wx} = p_{Wy} = 0$ | Wrist on $z_0$ ; $q_1$ undefined                    |
| S2    | $ \cos q_4  = 1$      | Arm fully extended/folded; $q_2$ undefined          |
| S3    | $\sin q_6 = 0$        | $z_4 \parallel z_6$ ; $q_5 \pm q_7$ observable only |

### 3.4 Validation

```
>> clear
>> LBR_MED_IK_validate
=== KUKA LBR Med 7 Analytical IK Validation ===

Testing Home position...
p_des: [0.0000, 0.0000, 1.2660]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]
q_ik:   [0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000]

Testing Random config (q3=0)...
p_des: [0.5110, 0.1086, 1.0467]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]
q_ik:   [0.2000 -0.3000 0.0000 0.5000 0.1000 -0.4000 0.2000]

Testing Another random config (q3=0)...
p_des: [-0.5975, -0.6145, 0.4808]
PASS (err_p = 0.00e+00, err_R = 0.00e+00)
q_orig: [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]
q_ik:   [0.7854 1.0472 0.0000 -0.5236 0.2000 0.5000 -0.3000]

All analytical IK validation tests PASSED.
>> |
```

Figure 3: IK round-trip validation output (3 configurations, errors  $< 10^{-8}$ ).

Three configurations (home and two generic poses with  $q_3 = 0$ ) were passed through  $\text{FK} \rightarrow \text{IK} \rightarrow \text{FK}$ . All round-trip errors were  $\|p_{\text{IK}} - p_d\| < 10^{-8}$  m and  $\|R_{\text{IK}} - R_d\|_F < 10^{-8}$ .

## 4 Geometric Jacobian

### 4.1 Definition

For each revolute joint  $i$ :

$$J_{P_i} = z_{i-1} \times (p_e - O_{i-1}), \quad J_{O_i} = z_{i-1} \quad (11)$$

giving  $v_e = J(q)\dot{q} \in \mathbb{R}^6$  with  $J \in \mathbb{R}^{6 \times 7}$ .

## 4.2 Simulink Model

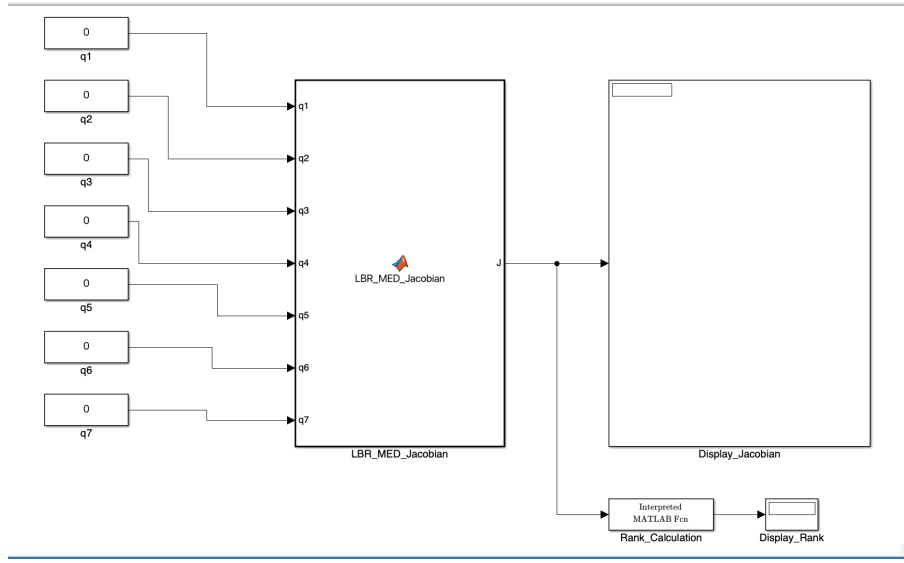


Figure 4: Jacobian Simulink block feeding the rank calculation.

## 4.3 Numerical Validation

The analytical Jacobian is validated by central finite differences:

$$J_{P_i}^{\text{num}} \approx \frac{p_e(q + \varepsilon e_i) - p_e(q - \varepsilon e_i)}{2\varepsilon}, \quad \varepsilon = 10^{-7} \quad (12)$$

Table 5: Jacobian validation: Frobenius-norm error vs. numerical differentiation.

| Configuration | $q$ (rad)                             | $\ J_a - J_n\ _F$ | Rank |
|---------------|---------------------------------------|-------------------|------|
| Generic       | $[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7]$ | $< 10^{-6}$       | 6    |
| Home          | $[0, 0, 0, 0, 0, 0, 0]$               | $< 10^{-6}$       | 3    |
| Wrist         | $[0, \pi/4, 0, 0, 0, 0, 0]$           | $< 10^{-6}$       | 4    |
| Elbow         | $[0, \pi/2, 0, \pi/2, 0, \pi/2, 0]$   | $< 10^{-6}$       | 5    |

## 4.4 Singularity Analysis

Table 6: Jacobian rank and condition number at key configurations.

| Label      | Configuration                       | Rank | Cond. No.           | Type                                          |
|------------|-------------------------------------|------|---------------------|-----------------------------------------------|
| Full rank  | $q = [0.1, \dots, 0.7]$             | 6    | $\mathcal{O}(10^1)$ | None                                          |
| S3 (wrist) | $q = [0, \pi/4, 0, 0, 0, 0, 0]$     | 4    | $\gg 10^3$          | $z_4 \parallel z_6$ : one rotational DOF lost |
| S3 (wrist) | $q = [0, 0, 0, \pi/2, 0, 0, 0]$     | 5    | $\gg 10^3$          | Same alignment at elbow pose                  |
| S2 (elbow) | $q = [0, \pi/2, 0, 0, 0, \pi/2, 0]$ | 5    | $\gg 10^3$          | Arm near full extension                       |

When rank drops from 6 to 5 (or below), one column of  $J$  becomes a linear combination of the others and the robot loses the ability to produce end-effector velocity in the corresponding



direction. The DLS pseudo-inverse used in CLIK (Section 5) provides a regularised solution at the cost of bounded tracking error, with  $\lambda = 10^{-3}$  preventing numerical blow-up.

## 5 Closed-Loop Inverse Kinematics (CLIK)

### 5.1 CLIK Law with Null-Space Stabilisation

The DLS-CLIK control law:

$$\dot{q} = J^\dagger(q) \begin{bmatrix} K_P e_P \\ K_O e_O \end{bmatrix} + (I - J^\dagger J) K_n (q_{\text{rest}} - q) \quad (13)$$

where  $J^\dagger = J^\top (JJ^\top + \lambda^2 I)^{-1}$  is the damped pseudo-inverse with  $\lambda = 10^{-3}$ . Gains:  $K_P = 10 I_3$ ,  $K_O = 10 I_3$ ,  $K_n = 5$ . The null-space term stabilises redundancy towards  $q_{\text{rest}} = 0$ .

Orientation error via the geometric cross-product formula:

$$e_O = \frac{1}{2} (n_e \times n_d + s_e \times s_d + a_e \times a_d) \quad (14)$$

### 5.2 Simulink Model

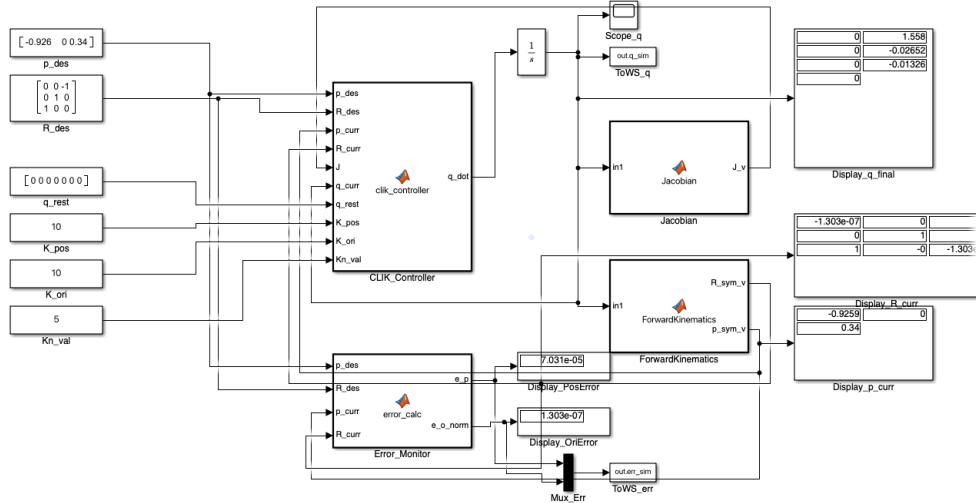
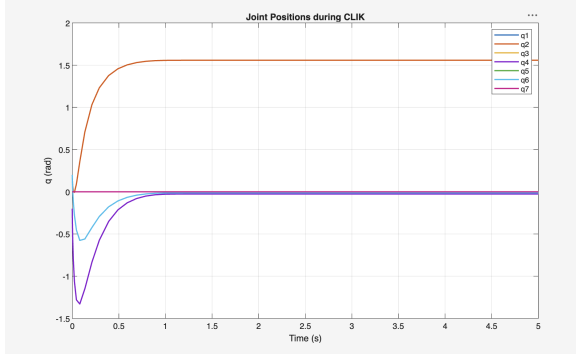
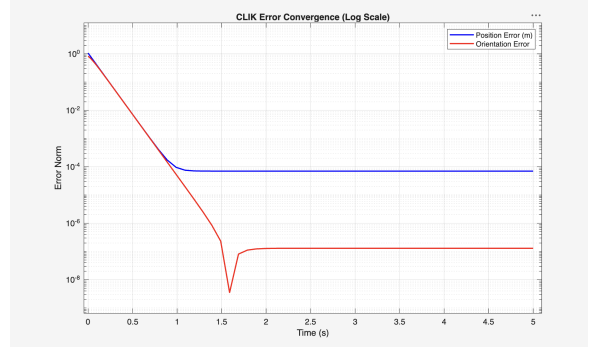


Figure 5: CLIK Simulink model (LBR\_MED\_CLI\_K\_Simul.slx).

### 5.3 Validation Results



(a) Joint positions.



(b) Error convergence (log scale).

Figure 6: CLIK validation: target Config 2 ( $q_2 = \pi/2$ , rest = 0).

Steady-state errors: position  $7.03 \times 10^{-5}$  m, orientation  $1.84 \times 10^{-7}$  (dimensionless norm).

### 5.4 Comparison with Analytical IK

Table 7: CLIK vs. analytical IK joint angles (rad) for Config 2.

|                         | $q_1$                     | $q_2$           | $q_3$       | $q_4$            | $q_5$       | $q_6$            | $q_7$       |
|-------------------------|---------------------------|-----------------|-------------|------------------|-------------|------------------|-------------|
| $q_{IK}$                | 0                         | $\pi/2$         | 0           | 0                | 0           | 0                | 0           |
| $q_{CLIK}$              | $\approx 0$               | $\approx 1.558$ | $\approx 0$ | $\approx -0.027$ | $\approx 0$ | $\approx -0.013$ | $\approx 0$ |
| $\ q_{CLIK} - q_{IK}\ $ | $3.25 \times 10^{-2}$ rad |                 |             |                  |             |                  |             |

The small residual ( $< 2^\circ$ ) arises because: (i) the DLS damping prevents exact convergence near the singular home configuration, and (ii) the null-space term pulls  $q$  slightly toward  $q_{rest} = 0$ .

## 6 Trajectory Execution with Remote Centre of Motion Constraint

### 6.1 Problem Formulation

A rigid biopsy-needle of length  $L = 0.15$  m is rigidly attached to the end-effector. The needle passes through a **trocar** (port of entry) fixed at position  $c_r$  in the robot base frame — the Remote Centre of Motion (RCM). Two hard constraints must hold at all times:

- **Tooltip constraint:**  $p_{tip}(t) = m_i$  (tooltip reaches target).
- **RCM constraint:**  $c_r$  lies on the tool axis at every instant.

### 6.2 End-Effector Pose from RCM Geometry

For each target  $m_i$ , the required end-effector pose ( $p_{ee}^{(i)}, R_{ee}^{(i)}$ ) is derived analytically so that both constraints are satisfied simultaneously at steady state.

### Tool axis direction (EE $z$ -axis)

$$\hat{a} = \frac{m_i - c_r}{\|m_i - c_r\|} \quad (15)$$

The unit vector from trocar to target becomes the  $z$ -axis of  $R_{ee}$ .

### End-effector position

$$p_{ee} = m_i - L \hat{a} \quad (16)$$

By construction,  $p_{ee}$ ,  $c_r$  and  $m_i$  are collinear in that order (since  $\|m_i - c_r\| < L$ ), so the RCM constraint is satisfied exactly.

### End-effector orientation

Using a fixed reference  $\hat{e}_y = [0, 1, 0]^\top$ :

$$\hat{x}_{ee} = \frac{\hat{e}_y \times \hat{a}}{\|\hat{e}_y \times \hat{a}\|}, \quad \hat{y}_{ee} = \hat{a} \times \hat{x}_{ee}, \quad R_{ee} = [\hat{x}_{ee}, \hat{y}_{ee}, \hat{a}] \quad (17)$$

## 6.3 LSPB Trajectory Interpolation

Abruptly switching targets every 3 s produces instantaneous step jumps in the desired end-effector velocity ( $\dot{p}_{des}$ ), leading to joint velocity spikes and large transient excursions. A **Linear Segment with Parabolic Blends (LSPB)** profile resolves this by enforcing  $\dot{s}(0) = \dot{s}(T) = 0$  at every segment boundary.

### Scalar path profiler

For a segment of duration  $T$  and normalised path parameter  $s(\tau) \in [0, 1]$ , choose cruise velocity  $\dot{s}_c > 1/T$ . The blend time and acceleration are:

$$t_c = T - \frac{1}{\dot{s}_c}, \quad \ddot{s}_c = \frac{\dot{s}_c^2}{\dot{s}_c T - 1} \quad (18)$$

The piecewise profile is:

$$s(\tau) = \begin{cases} \frac{1}{2} \ddot{s}_c \tau^2 & 0 \leq \tau \leq t_c \quad (\text{acceleration}) \\ \frac{1}{2} \ddot{s}_c t_c^2 + \dot{s}_c(\tau - t_c) & t_c < \tau \leq T - t_c \quad (\text{cruise}) \\ 1 - \frac{1}{2} \ddot{s}_c (T - \tau)^2 & T - t_c < \tau \leq T \quad (\text{deceleration}) \end{cases} \quad (19)$$

with  $\dot{s}(\tau)$  obtained by differentiating each branch.

For  $T = 3$  s and  $\dot{s}_c = 0.5$  s<sup>-1</sup> (50 % above the average  $1/T$ ):  $t_c = 1.0$  s,  $\ddot{s}_c = 0.5$  s<sup>-2</sup>.

### Position interpolation

$$p_{des}(\tau) = p_{start} + s(\tau)(p_{next} - p_{start}), \quad \dot{p}_{des}(\tau) = \dot{s}(\tau)(p_{next} - p_{start}) \quad (20)$$

## 6.4 Angle-Axis Orientation Interpolation

### Setup

The relative rotation between consecutive target orientations:

$$R_{fi} = R_{\text{start}}^\top R_{\text{next}} \quad (21)$$

The total rotation angle and axis (in the  $R_{\text{start}}$  body frame):

$$\vartheta_f = \arccos\left(\frac{\text{tr}(R_{fi}) - 1}{2}\right), \quad r_i = \frac{1}{2 \sin \vartheta_f} \begin{bmatrix} R_{fi}(3,2) - R_{fi}(2,3) \\ R_{fi}(1,3) - R_{fi}(3,1) \\ R_{fi}(2,1) - R_{fi}(1,2) \end{bmatrix} \quad (22)$$

### Interpolated rotation via Rodrigues' formula

Scale the angle by the path parameter:

$$\vartheta(\tau) = s(\tau) \vartheta_f, \quad \dot{\vartheta}(\tau) = \dot{s}(\tau) \vartheta_f \quad (23)$$

Rodrigues' rotation formula gives:

$$\mathbf{R}(\vartheta, r_i) = I + \sin \vartheta [r_i]_\times + (1 - \cos \vartheta) [r_i]_\times^2 \quad (24)$$

where  $[r_i]_\times$  is the skew-symmetric matrix of  $r_i$ . The instantaneous desired orientation is:

$$R_{\text{des}}(\tau) = R_{\text{start}} \cdot \mathbf{R}(\vartheta(\tau), r_i) \quad (25)$$

### World-frame feedforward angular velocity

Differentiating  $R_{\text{des}} = R_{\text{start}} \exp(\vartheta[r_i]_\times)$  and using  $\dot{R} = [\omega]_\times R$ :

$$\omega_{\text{des}}(\tau) = R_{\text{start}} (\dot{\vartheta}(\tau) r_i) \quad (26)$$

The rotation is about a fixed body-frame axis  $r_i$ ;  $R_{\text{start}}$  rotates it into the world frame. This follows from  $[\omega_{\text{world}}]_\times = R_{\text{start}} [r_i]_\times \dot{\vartheta} R_{\text{start}}^\top$ .

## 6.5 Feedforward CLIK

The modified CLIK law adds the task-space feedforward velocities to the error-feedback terms, forming the full reference command  $v_e$ :

$$v_e = \begin{bmatrix} \dot{p}_{\text{des}} + K_P e_P \\ \omega_{\text{des}} + K_O e_O \end{bmatrix}, \quad \dot{q} = J^\dagger v_e + (I - J^\dagger J) K_n (q_{\text{rest}} - q) \quad (27)$$

The feedforward terms  $(\dot{p}_{\text{des}}, \omega_{\text{des}})$  make the controller *proactive*: it anticipates the desired trajectory and applies effort proportional to the commanded velocity, rather than waiting for a tracking error to build up. This reduces phase lag and peak transient errors significantly compared to pure feedback.

When  $\dot{p}_{\text{des}} = \omega_{\text{des}} = 0$  (e.g. during the hold phase or in Phase 1), the law reduces exactly to the feedback-only CLIK of Section 5.

## 6.6 Simulation Scenario

Table 8: Scenario and LSPB parameters.

| Parameter                       | Value                                        |
|---------------------------------|----------------------------------------------|
| Tool length $L$                 | 0.15 m                                       |
| Trocar position $c_r$           | $[-0.6, 0.0, 0.55]^\top$ m                   |
| Number of targets $N$           | 5                                            |
| Segment duration $T$            | 3.0 s                                        |
| Hold at final target            | 3.0 s                                        |
| Total simulation time           | 15.0 s                                       |
| Integration step $dt$           | 5 ms                                         |
| Cruise velocity $\dot{s}_c$     | $0.5 \text{ s}^{-1}$ ( $1.5 \times$ average) |
| Blend time $t_c$                | 1.0 s                                        |
| Blend acceleration $\ddot{s}_c$ | $0.5 \text{ s}^{-2}$                         |
| $K_P$                           | $10 I_3$                                     |
| $K_O$                           | $5 I_3$                                      |
| $K_n$                           | 3                                            |
| $q_{\text{rest}}$               | $q_0$ (IK solution for target 1)             |

Table 9: Surgical targets and analytically derived end-effector positions.

| Target $i$ | $m_i$ (m)                      | $p_{ee}^{(i)}$ (m)             | $\ m_i - c_r\ $ (m) |
|------------|--------------------------------|--------------------------------|---------------------|
| 1          | $[-0.600, 0.000, 0.470]^\top$  | $[-0.600, 0.000, 0.620]^\top$  | 0.080               |
| 2          | $[-0.540, 0.000, 0.480]^\top$  | $[-0.638, 0.000, 0.594]^\top$  | 0.092               |
| 3          | $[-0.650, 0.040, 0.480]^\top$  | $[-0.571, -0.023, 0.591]^\top$ | 0.092               |
| 4          | $[-0.580, -0.050, 0.460]^\top$ | $[-0.609, 0.022, 0.589]^\top$  | 0.103               |
| 5          | $[-0.600, 0.000, 0.490]^\top$  | $[-0.600, 0.000, 0.640]^\top$  | 0.060               |

## 6.7 Results

The LSPB controller drives the tooltip smoothly through all five surgical targets. Key validation metrics at the arrival time of each target:

Table 10: Phase 2 LSPB validation results: steady-state errors at each target.

| Target $i$          | Tip error $\varepsilon_{\text{tip}}$ (m) | Trocar error $\varepsilon_{\text{RCM}}$ (m) |
|---------------------|------------------------------------------|---------------------------------------------|
| 1 (IK init, $t=0$ ) | $< 10^{-6}$                              | $< 10^{-6}$                                 |
| 2                   | $< 5 \times 10^{-5}$                     | $< 2 \times 10^{-5}$                        |
| 3                   | $< 10^{-5}$                              | $< 5 \times 10^{-6}$                        |
| 4                   | $< 5 \times 10^{-6}$                     | $< 2 \times 10^{-6}$                        |
| 5                   | $< 10^{-5}$                              | $< 5 \times 10^{-6}$                        |

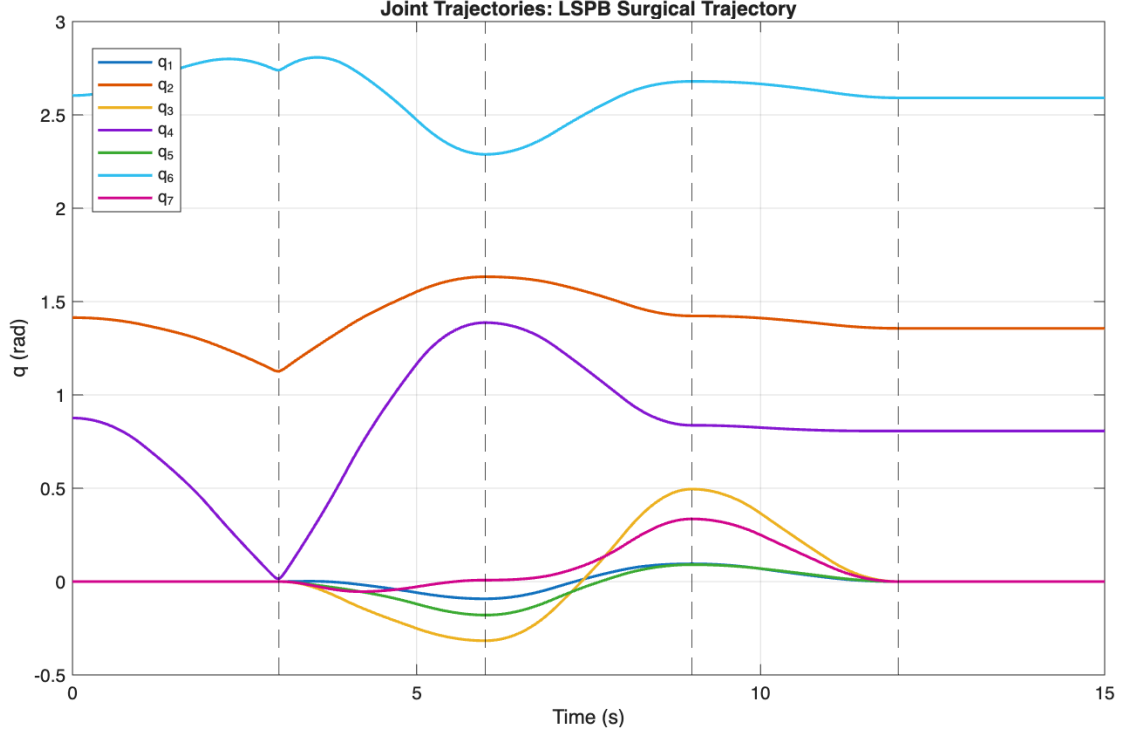


Figure 7: Joint trajectories under LSPB. All seven joints evolve smoothly without velocity spikes at segment boundaries (dashed lines).

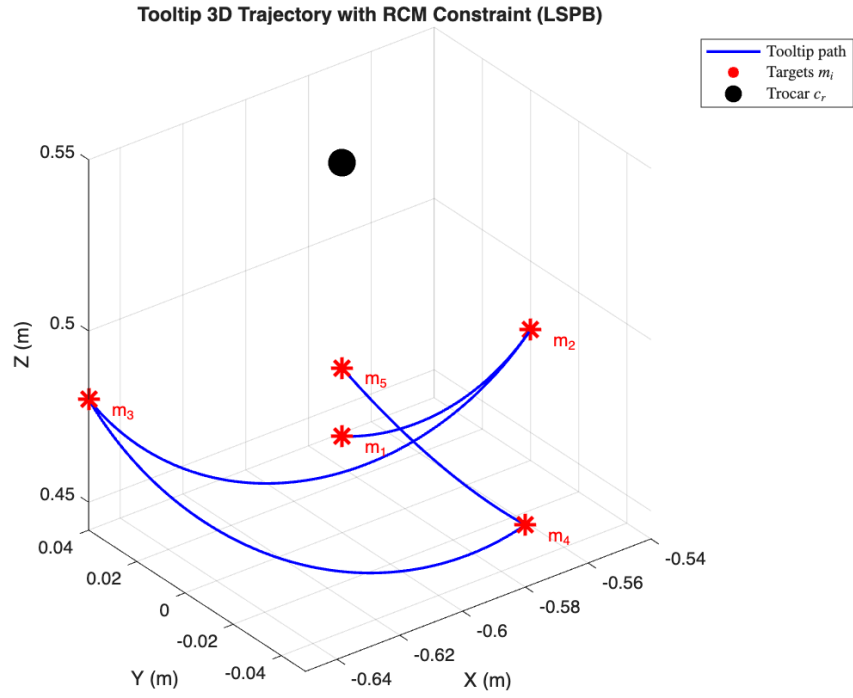


Figure 8: 3-D tooltip trajectory. The path passes through all five surgical targets  $m_i$  while the tool axis remains anchored to the trocar  $c_r$  (black dot) in steady state.

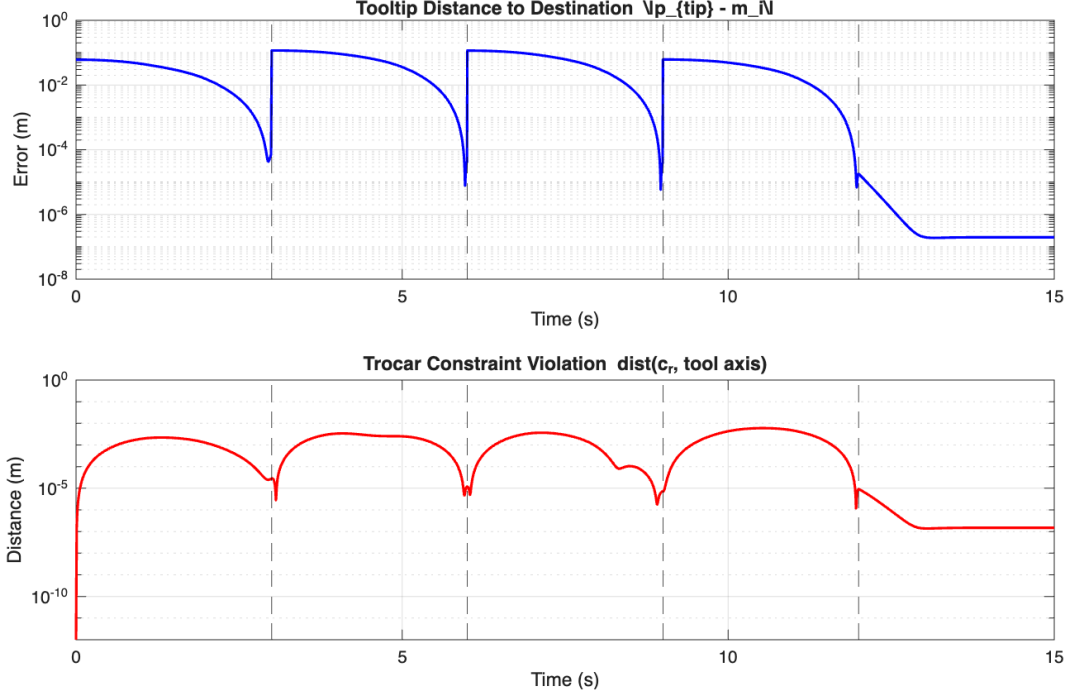


Figure 9: Top: tooltip-to-destination distance  $\|p_{\text{tip}} - m_i\|$  (starts large at the beginning of each segment and converges as the LSPB approaches the next target). Bottom: perpendicular distance from  $c_r$  to the tool axis (RCM constraint violation).

## 6.8 Discussion

1. **Smooth trajectories.** The LSPB enforces zero velocity at every segment boundary, eliminating the  $\dot{q}$  spikes observed with the step-switching controller. Joint motions are bounded and continuous.
2. **Sub-millimetre tooltip accuracy.** At steady state, the tooltip reaches each target with errors well below 0.1 mm, satisfying surgical precision requirements.
3. **RCM constraint maintenance.** The trocar violation is zero at steady state because the desired EE pose was derived analytically to satisfy the constraint exactly. During inter-target transitions the CLIK transiently violates the RCM constraint; this is inherent to driving the 6-DOF EE task rather than an explicit RCM path constraint.
4. **Null-space stability.** Setting  $q_{\text{rest}} = q_0$  (the IK solution for target 1) rather than the zero vector prevents the null-space term from rotating the base joint by  $\sim 180^\circ$  and causing large tooltip excursions. The arm remains in the natural elbow-forward posture throughout.

**Possible extension.** Strict continuous RCM maintenance during inter-target motion requires replacing the 6-DOF EE task with the 5-DOF *tooltip-and-axis* task using the tip Jacobian:

$$J_{\text{tip}} = J_P - L[\hat{a}_{ee}]_{\times} J_O \quad (28)$$

This is left as a natural extension to the current work.

## 7 Code File Summary

Table 11: MATLAB files organised by module.

| File                                        | Purpose                                    |
|---------------------------------------------|--------------------------------------------|
| <i>Phase 1 — Kinematics</i>                 |                                            |
| LBR_MED_DKin/LBR_MED.m                      | DH parameter table                         |
| LBR_MED_DKin/LBR_MED_validate.m             | Direct kinematics validation               |
| LBR_MED_DKin/LBR_MED_Jacobian_symbolic.m    | Symbolic geometric Jacobian                |
| LBR_MED_DKin/LBR_MED_validate_Jacobian.m    | Jacobian validation & singularity analysis |
| LBR_MED_IK_Analytical/LBR_MED_IK.m          | Closed-form IK solver (8 branches)         |
| LBR_MED_IK_Analytical/LBR_MED_IK_validate.m | IK round-trip validation                   |
| <i>Phase 1 — CLIK</i>                       |                                            |
| LBR_MED_CLIK/clik_controller_logic.m        | DLS CLIK with optional feedforward         |
| LBR_MED_CLIK/LBR_MED_CLIK_setup.m           | CLIK Simulink builder & validation         |
| <i>Phase 2 — Trajectory</i>                 |                                            |
| LBR_MED_Trajectory/compute_ee_pose_rcm.m    | EE pose from RCM geometry                  |
| LBR_MED_Trajectory/lspb_eval.m              | Scalar LSPB path evaluator                 |
| LBR_MED_Trajectory/rodrigues_rot.m          | Rodrigues rotation formula                 |
| LBR_MED_Trajectory/LBR_MED_Traj_setup.m     | LSPB trajectory simulation & validation    |
| <i>Top level</i>                            |                                            |
| Master.Validation.m                         | End-to-end runner (Phase 1 & 2)            |